Academic Assignment: Multi-Audience Communication, Multi-Metric Optimization, and Empirical Reporting

Coursework Submission | TrendPulse AI Evaluation

Question 1

Universal Overview

TrendPulse AI is an automated financial intelligence platform that forecasts short-term equity price momentum and volatility by fusing tick-level order book dynamics with streaming market news sentiment. It translates algorithmic risk signals into disciplined, transparent portfolio guidance, enabling users to manage downside exposure and make data-driven investment decisions.

Epistemic Categorization

- **Documented Facts:** Financial time series exhibit extreme non-stationarity, leptokurtic return distributions, and volatility clustering (Mandelbrot, 1963; Engle, 1982). Standard integer differencing eliminates non-stationarity but erases multi-period memory, whereas fractional differentiation preserves long-range temporal dependence (López de Prado, 2018).
- **Operating Assumptions:** Microblogging feeds and regulatory disclosure feeds contain exploitable directional sentiment signals that precede institutional price discovery by several seconds to minutes.
- **Proposed System Architecture:** A dual-stream deep neural network coupling a Temporal Fusion Transformer (TFT) for high-frequency price history with a fine-tuned, domain-adapted FinBERT model for streaming textual sentiment, unified via a cross-attention layer and optimized against an asymmetric downside-risk loss function.

Solution 1: Executive Explanation for Technical Researchers

TrendPulse AI is a multi-modal quantitative architecture engineered to forecast directional equity returns and conditional volatility regimes under non-stationary market dynamics. To resolve the memory-loss dilemma inherent to standard integer differencing, the data ingestion pipeline applies fractional differentiation ($d \approx 0.35$), ensuring stationarity while retaining long-term temporal autocorrelation.

The inference engine utilizes a dual-stream architecture. The quantitative pathway deploys a Temporal Fusion Transformer (TFT) with multi-head self-attention to process asynchronous limit order book snapshots, historical tick volumes, and cross-asset volatility indices (VIX). Concurrently, the qualitative pathway runs a quantized, domain-specific FinBERT model that parses

unstructured corporate filings, real-time earnings transcripts, and financial wire feeds into calibrated dense sentiment vectors. These representations coalesce within a multi-headed cross-attention fusion layer, optimized via an asymmetric Sortino-ratio loss function that penalizes downside semivariance rather than symmetric volatility.

To eliminate look-ahead bias and cross-validation leakage, the evaluation harness utilizes combinatorial purged cross-validation paired with an embargo interval. Validated across a ten-year out-of-sample window on S&P 500 constituents, the architecture demonstrates an annualized Sharpe ratio of 1.84 with a maximum drawdown under 9.2%, establishing robust generalization across distinct market regimes over standard GARCH and recurrent baselines.

Solution 2: Executive Explanation for University Management

TrendPulse AI establishes an institutional capability to strengthen endowment capital preservation while generating disciplined, risk-adjusted returns. Modern university endowments face a dual mandate: funding ongoing academic operational commitments—such as student financial aid, endowed faculty chairs, and research facilities—while complying with strict fiduciary standards under the Uniform Prudent Management of Institutional Funds Act (UPMIFA).

Traditional endowment management allocates heavily to external hedge funds and active asset managers, incurring 1% to 2% base management fees alongside 20% performance fees that erode compound growth over generational horizons. TrendPulse AI serves as an internal, automated risk-management platform for the university treasury. Rather than pursuing speculative high-frequency trading, the platform emphasizes automated downside capital protection. It continuously monitors macroeconomic indicators, sector trends, and news sentiment, delivering pre-emptive alerts that allow treasury officers to de-risk equity exposure ahead of broader market corrections.

In backtested portfolio stress tests covering historical market downturns, this systematic rebalancing preserved principal capital and generated a 2.3% net annual yield advantage over static benchmark indices. Deployed entirely within the university's private institutional cloud, TrendPulse AI ensures complete data provenance, governance, and auditability without exposing institutional funds to third-party management overhead.

Solution 3: Executive Explanation for Ordinary Users

Managing personal investments often feels confusing, intimidating, and stressful. Most people want their hard-earned savings to grow safely, but they face endless financial jargon, sensationalized news headlines, and complex charts that make it difficult to know when to invest and when to hold back.

TrendPulse AI functions as a calm, personal financial co-pilot on your smartphone. Instead of requiring you to read lengthy quarterly financial reports or monitor ticker symbols all day, the app translates complex market activity into clear, plain-language guidance. When you open your dashboard, you are not confronted with complicated mathematical formulas or trading screens. Instead, you receive straightforward, color-coded status cards and actionable summaries, such as: *"Market volatility is high this week ahead of central bank interest rate announce-*

ments. We recommend maintaining your cash reserves rather than purchasing new technology stocks today."

TrendPulse AI is not a get-rich-quick scheme or an automated bot that trades your money without your knowledge. Its primary purpose is to safeguard your savings against emotional mistakes, such as panic-selling during temporary market declines. You retain complete ownership and control over every deposit, withdrawal, and trade, supported by objective, reliable intelligence.

Question 2

Hypothetical Experimental Results Baseline

Table 1: Empirical Model Performance and Resource Consumption

Method	Accuracy (%)	Training Time (min)	Memory Usage
Method A	91%	18 min	420 MB
Method B	94%	41 min	850 MB
Method C	93%	22 min	510 MB
Method D	95%	76 min	1.2 GB

1. Recommended Visualization Strategy

To capture the multi-criteria trade-offs among Accuracy, Training Time, and Memory Usage, a single-variable plot is analytically insufficient. The optimal approach employs a **combination of two complementary visualizations** grounded in multi-objective optimization literature (Deb, 2001; Tufte, 2001):

- **Primary Visualization: Multi-Attribute Pareto Frontier Scatter Plot (Bubble Plot)**
  X-axis: Training Time in minutes (computational latency; minimizing direction).
  Y-axis: Classification Accuracy as a percentage (predictive efficacy; maximizing direction).
  Bubble Area: Memory Usage in Megabytes (RAM footprint; minimizing direction).
  Visual Annotation: A step-wise empirical Pareto frontier line connecting non-dominated solutions (Method A $\rightarrow$ Method C $\rightarrow$ Method D). This explicitly exposes Method B as an inefficient, dominated candidate falling well below the efficiency curve.
- **Secondary Visualization: Multi-Panel Faceted Dot / Lollipop Plot**
  A set of horizontally aligned panels displaying Accuracy (%), Training Time (min), and Memory (MB) side-by-side for Methods A through D. Each metric retains its true native scale and physical units, preventing optical distortion caused by forced normalization or secondary axes.

2. Inadequacy of a Single Bar Chart

A single bar chart provides an incomplete and structurally biased interpretation of machine learning model evaluations for two critical reasons:

1. **Dimensional Omission and Selection Bias:** A conventional bar chart can display only one dependent variable across discrete categories (most often chosen as Accuracy). When plotted alone, Method D (95%) appears as the undisputed winner, concealing that it requires more than four times the training duration of Method A (76 min vs. 18 min) and nearly triple its memory footprint (1.2 GB vs. 420 MB).

2. **Incompatible Units and Metric Baselines:** Accuracy is a bounded ratio metric ($[0\%, 100\%]$) where higher values represent superior performance. Conversely, training latency (minutes) and memory consumption (megabytes) are unbounded physical consumption metrics where lower values are desirable. Forcing these heterogeneous dimensions onto a single shared vertical axis requires arbitrary scaling or normalization coefficients, distorting the true relative cost of marginal performance gains.

3. Multi-Metric Trade-off Analysis

The empirical data demonstrates the principle of **diminishing marginal returns** and the computational cost scaling highlighted in Green AI research (Schwartz et al., 2020):

- **The Efficient Step (Method A $\rightarrow$ Method C):** Transitioning from Method A to Method C yields an absolute $+2.0\%$ improvement in predictive accuracy (91% to 93%) at a minor resource cost of only 4 additional minutes ($+22.2\%$) and 90 MB of memory ($+21.4\%$). This represents a highly favorable marginal trade-off.
- **The Inefficient Outlier (Method C $\rightarrow$ Method B):** Method B gains only $+1.0\%$ in accuracy over Method C (93% to 94%), but requires an 86.4% surge in training time (22 min to 41 min) and a 66.7% expansion in memory footprint (510 MB to 850 MB). Method B represents an inefficient allocation of compute.
- **The High-Cost Ceiling (Method C $\rightarrow$ Method D):** Securing the top accuracy of 95% in Method D ($+2.0\%$ over Method C) triggers a severe super-linear cost penalty: training time escalates by $+245.5\%$ (22 min to 76 min) and memory allocation expands by $+135.3\%$ (510 MB to $1,200$ MB).

Analytical Verdict: Method C represents the operational "knee point" (sweet spot) of the Pareto frontier, capturing near-optimal accuracy while maintaining a lean computational profile.

4. Low-Resource and Edge Device Deployment Scenarios

If this model were targeted for edge execution (e.g., an ARM Cortex microcontroller, mobile APU, Raspberry Pi gateway, or automotive electronic control unit), the operational constraints shift from accuracy-first to strict physical resource feasibility:

1. **Immediate Elimination of Method D:** Low-resource devices typically operate with hard physical RAM ceilings (e.g., 512 MB to 1 GB total unified system memory). Method D's 1.2 GB requirement exceeds these physical boundaries entirely, triggering operating system kernel panics, Out-Of-Memory (OOM) process termination, or severe flash memory wear from disk swapping.
2. **Thermal Dissipation and Battery Drain:** In edge devices operating without active cooling, extended computation time translates directly into thermal throttling and rapid battery exhaustion. Method D's 76 minutes of training latency is prohibitive for on-device incremental learning.

3. **Revised Operational Ranking:**
    - If hardware RAM is capped at 512 MB, **Method A (**420 **MB,** 18 **min)** is the **sole viable deployment candidate**, maintaining an 18% memory buffer (92 MB) for kernel and auxiliary tasks. Method C (510 MB) operates at the absolute brink of memory exhaustion.
    - If hardware RAM is 1 GB, **Method C (**510 **MB,** 22 **min)** becomes the superior selection, securing 93% accuracy while preserving roughly 500 MB of memory headroom.

5. Results and Discussion Section

Empirical evaluation reveals that while Method D attained the highest classification accuracy (95%), it introduces prohibitive computational overhead, requiring 76 minutes of training and 1.2 GB of memory. Conversely, Method A demonstrated minimal resource utilization (18 minutes, 420 MB) but lagged in predictive efficacy (91%). Method C established the empirical Pareto-optimal configuration, achieving 93% accuracy within 22 minutes and consuming 510 MB. In contrast, Method B displayed severe marginal inefficiency: capturing an incremental 1% accuracy over Method C required an 86.4% increase in training duration (41 minutes) and a 66.7% expansion in memory footprint (850 MB). Furthermore, attaining Method D's marginal 2% gain over Method C demanded a 245.5% surge in training latency alongside more than double the memory allocation. Unless task requirements impose uncompromising precision thresholds where computational expenditure is entirely unconstrained, Method C represents the most robust, scalable candidate for production deployment, particularly in environments governed by bounded memory ceilings or strict thermal dissipation budgets.

6. Misleading Presentation Strategy & Critical Analysis

- **Misleading Presentation Technique:** A **truncated vertical axis bar chart** displaying exclusively the Accuracy metric, with the vertical axis scaled narrowly from 90% to 95.5%, while completely omitting Training Time and Memory Usage from the visual presentation.
- **The Optical Illusion Created:** By truncating the baseline at 90% instead of 0%, the visual bar height difference between Method A (91%) and Method D (95%) is optically magnified by a factor of five. This gives non-technical stakeholders the impression that Method D is vastly superior, when in reality the absolute performance gain is merely 4.0 percentage points.
- **Deliberate Concealment of Operational Costs:** Suppressing the time and memory dimensions hides the reality that Method D requires a 322% increase in training time and a 186% increase in memory over Method A. Stakeholders viewing this truncated chart would approve Method D without knowing it imposes exponential infrastructure costs and latency bottlenecks.

Question 3

Context & Research Hypothesis

Within the **TrendPulse AI** quantitative finance framework established in Question 1, the core empirical research hypothesis was formulated as follows:

- **Primary Hypothesis (H_1):** Integrating real-time streaming financial news and social sentiment embeddings via a multi-head cross-attention layer into a limit order book Temporal Fusion Transformer (TFT) will yield a statistically significant improvement ($p < 0.01$) in 5-minute intraday directional return prediction accuracy and out-of-sample portfolio Sharpe ratio over an order-book-only unimodal baseline.

Following rigorous backtesting on tick-level out-of-sample S&P 500 equity data, the empirical results fell noticeably below expectations, and H_1 **was not supported by the data**. The following section provides the formal research report communicating this outcome.

Structure of the Research Report Section

1. Results

Experimental evaluation across 2.4 million out-of-sample 5-minute trade intervals demonstrated that the proposed multimodal architecture (TFT + FinBERT cross-attention) failed to outperform the unimodal numerical baseline (TFT price/volume alone):

- **Directional Accuracy:** The unimodal TFT baseline achieved a directional accuracy of 54.12% (SD $= 0.41\%$), whereas the multimodal model achieved 51.08% (SD $= 0.63\%$). A paired Diebold-Mariano test confirmed that this performance divergence is statistically significant ($t = -3.84$, $p = 0.00012$), indicating an active degradation in predictive performance upon adding the textual stream.
- **Risk-Adjusted Return:** Under simulated execution with standard institutional transaction fees (5 bps per trade), the unimodal baseline achieved an annualized Sharpe ratio of 1.82 with a maximum drawdown of 8.9%. Conversely, the multimodal model produced an annualized Sharpe ratio of 0.74 with a maximum drawdown of 18.4%.
- **Calibration:** The Expected Calibration Error (ECE) rose from 0.028 in the unimodal configuration to 0.089 in the multimodal model, showing that textual sentiment embeddings introduced substantial overconfidence in probability outputs.

2. Interpretation

These findings lead directly to the rejection of Hypothesis H_1. Rather than delivering additive orthogonal predictive alpha, incorporating the high-dimensional streaming sentiment vector

(768-dimensional FinBERT dense embeddings) introduced high-variance noise into the attention mechanism. At a 5-minute forecasting granularity, the model overfitted to transient semantic fluctuations. The cross-attention fusion weights assigned disproportionate importance to sensationalized or ambiguous news headlines, which diluted the higher-frequency, structurally predictive microstructural signals derived from order book bid-ask spreads and depth imbalances.

3. Comparison with Previous Literature

Our results appear to contradict foundational literature advocating sentiment-driven equity forecasting (e.g., Bollen et al., 2011; Araci, 2019), which reported predictive gains ranging from 2% to 6% when integrating textual sentiment into market models.

However, this divergence aligns precisely with financial microstructure literature (Groß-Klußmann & Hautsch, 2011; Boudoukh et al., 2019). Those studies emphasize that while textual sentiment exhibits statistically significant explanatory power at daily, weekly, or multi-hour aggregation windows, its predictive efficacy collapses at sub-hourly horizons. In modern electronic markets, institutional algorithmic market makers price public news announcements into order books within tens to hundreds of milliseconds. Consequently, by the time public text feeds are ingested, tokenized, and embedded by our transformer pipeline (1.2 seconds average latency), the informational advantage has vanished, leaving only residual market noise.

4. Possible Reasons for the Unexpected Result

1. **Temporal Latency Mismatch:** Quantitative order book features update synchronously at millisecond intervals, whereas news events arrive asynchronously and intermittently. Forcing the cross-attention layer to align continuous numerical ticks with sparse, lagging text tokens forced the model to interpolate embeddings during news-sparse intervals, generating phantom correlations.
2. **Feature Space Asymmetry and Gradient Domination:** The textual embedding space (768 continuous dimensions per token) heavily outnumbered the lower-dimensional engineered order book features (32 engineered variables). During backpropagation, loss gradients from the cross-attention mechanism dominated weight updates, destabilizing the trained temporal representations of the TFT backbone.
3. **Semantic Ambiguity and Sarcasm:** In-depth error profiling revealed that FinBERT frequently misclassified sarcastic retail commentary and ambiguous regulatory disclosures as strongly directional, inducing high-confidence false breakout trades.

5. Limitations

- **Fixed Granularity:** The study evaluated predictions strictly at an invariant 5-minute horizon; it did not assess whether textual features recover predictive efficacy at 30-minute, 1-hour, or end-of-day horizons.

- **Static Fusion Weights:** The cross-attention architecture lacked an adaptive, learnable gating threshold to dynamically sever the textual pathway during periods of low sentiment confidence or news absence.
- **Feed Latency:** Text collection relied on web-socket financial APIs rather than collocated direct-market data feeds, introducing an unmeasured millisecond latency overhead.

6. Implications

- **Theoretical Implication:** Textual sentiment is not universally beneficial across temporal resolutions. A clear boundary condition exists: natural language representations degrade rather than enhance predictive power when applied to sub-hourly market forecasting.
- **Practical / Engineering Implication:** Removing the FinBERT stream eliminates the most computationally expensive component of TrendPulse AI. System inference latency decreases by 78% (from $1,240$ ms to 270 ms), memory footprint drops from 1.8 GB to 340 MB, and operational cloud hosting costs fall by 65%, yielding a leaner, faster, and more profitable production deployment.

7. Conclusion

Although the empirical results failed to validate H_1, the study produces an essential disconfirming insight: fine-grained textual sentiment fusion is structurally unsuited for high-frequency directional equity forecasting. Eliminating this component preserves model parsimony, lowers latency, and protects the system from semantic overconfidence.

8. Future Work

- **Asymmetric Multi-Horizon Conditioning:** Restructuring the architecture so that FinBERT embeddings are restricted to predicting multi-hour volatility regimes (VIX shifts), while the order book TFT operates unimodally for 5-minute directional momentum.
- **Dynamic Uncertainty Gating:** Designing an entropy-based gating module that disables the sentiment pathway whenever text classification entropy exceeds a calibrated threshold ($\mathcal{H} > 0.35$).

Critical Analysis: When Explaining Unexpected Results Strengthens vs. Weakens Research Credibility

Table 2: Epistemic Comparison: Strengthening Rigor vs. Methodological Discredit

Evaluation Dimension	When Explaining Strengthens the Report	When Explaining Weakens / Discredits Methodology
Scientific Integrity & Objectivity	**Transparent Negative Findings:** Honestly reporting that H_1 failed combats publication bias (the "file-drawer effect"; Rosenthal, 1979) and establishes high epistemic integrity.	**Post-Hoc Rationalization (HARKing):** Modifying hypotheses after results are known (Kerr, 1998) or presenting post-hoc correlations as a priori predictions erodes peer trust.
Diagnostic Rigor & Root-Cause Attribution	**Grounded Mechanistic Explanations:** Isolating specific, testable mechanisms—such as the Diebold-Mariano test showing feature interference and latency mismatch—demonstrates mastery.	**Vague Defensiveness & Hand-Waving:** Attributing failures to ambiguous excuses (e.g., *"the market was unusually irrational"*) signals a lack of analytical rigor.
Experimental Controls & Validation	**Validation of Robust Controls:** Explaining that purged cross-validation properly caught overfitting confirms that the testing harness was rigorous and protected against false positives.	**Concealing Methodological Flaws:** Attempting to excuse poor results when the true issue was uncalibrated hyperparameters or data leakage makes methodology appear defective.
System Refinement & Strategic Utility	**Actionable Engineering Boundary:** Showing that removing FinBERT cuts latency by 78% and infrastructure costs by 65% delivers immense practical utility to industry.	**Moving Goalposts:** Arbitrarily shifting performance metrics mid-stream to claim a false "win" reveals an unreliable, biased evaluation framework.

Concrete Examples from the TrendPulse AI Project

1. **How It Strengthens the Work (Methodological Rigor):**
   The research team performs an ablation analysis isolating the FinBERT pathway. By plotting feature contribution weights, they demonstrate that as news ingestion latency exceeds 500 milliseconds, the correlation between text sentiment and forward 5-minute price direction drops to zero ($r = 0.01$, $p = 0.88$). This empirical demonstration proves that the experiment was designed with sufficient sensitivity to detect signal decay. It transforms a "failed prediction" into an authoritative discovery regarding the temporal boundaries of NLP in algorithmic finance.

2. **How It Weakens the Work (Methodological Unreliability):**
   Upon observing that the multimodal model achieved only 51.08% accuracy, the authors write: *"The model would have achieved superior results, but market-makers manipulated price spreads, and the news sentiment API occasionally returned inaccurate timestamps."* This explanation relies on unverified external blame without diagnostic data. It implies that the researchers failed to clean their input timestamps, did not account for standard exchange

dynamics, and are attempting to excuse an inadequate experimental pipeline rather than diagnosing genuine algorithmic performance.